```
while (complement_num != 0) {
    rem3 = complement_num % 10;
    complement_num /= 10;
    ans3 += rem3 * multi3;
    multi3 *= 2;
}
cout << "complemented Integer Value: " << ans3 << endl;
return 0;
}
```

CONCEPT :-

To convert Binary to decimal at last, we cannot use typecasting

WRONG `cout << " complemented Integer value:" << (int) complement_num;`

it is because,

5 → 101 (binary)

complement → 010

complement_num which is an int type variable stores 10 of 010 as integer value, so typecasting won't work.

• FUNCTION: -

A function is a self-contained block of code that performs a specific task. It can take inputs (Parameters), execute statements, and optionally return a value (when return type is not void).

Syntax →

```
return_type function_name (Parameter 1, Parameter 2, ...) {
    Statements;
    return value;
}
```

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- **Default Parameter / Value:-**

A default value (default Parameter) is a value assigned to a Function that is automatically used if no argument is provided when the function is called.

ex →

```
int fact (int n=3){
        int fact = 1;
        for(int i=1 ; i<=n; i++){
                fact *=i;
        }
        return fact;
}
```

```
fact ();   → Uses default value n=3
fact (5);  → Uses n=5
```

- **Pass By Value:-**

In pass by value, a copy of the actual argument is passed to the function. Any changes made inside the function to the parameter donot effect the original variable.

ex →

```
void in (int n){
        n++;
}
int main(){
        int a=10;
        in (a);
        cout << a;   → Output = 10
}
```

- **Pass By Reference:-**

In pass by reference, the function parameter becomes a reference to the original variable. Any changes made inside the function directly affects the original variable.

ex →

```
void in (int &n) {
```

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• Function Overloading :-

Function overloading is a Feature that allows multiple Functions to have the Same name but different parameter list (different number, type, or order of parameters) -

ex →
```
#include <iostream>
using namespace std;
int sum(int a, int b) {
        return a+b;
}
float sum (float a, floatb){
        return a+b;
}
int main() {
    cout << "Sum:" << sum (10,20) << endl;   → 30
    cout << "Sum:" << sum(2.5f , 6.5f) << endl;  → 9
    cout << "Sum:"
}
```

• Swap Function:-

Swap() is a predefined function in C++ used to exchange the values of two variables.

ex →
```
int a=10, b=20;
swap(a,b);     → Then  a = 20 , b=10
```

• Square Root Function:-

sqrt() is a predefined function used to find the square root of a number.

ex → sqrt(2) ⇒ 1.41421.....

Q) WAP to create functions to check for prime number and to find the factorial.

```
#include <iostream>
using namespace std;
```

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```
bool prime (int num) {
    if (num <=1) {
        return 0
    }
    for (int i=2; i <num; i++) {
            if (num % i == 0) {
                return 0;
            }
    }
    return 1;
}

int Factorial (int num) {
    int fact =1;
    for (int i=2; i <= num; i++) {
            fact *=1;
    }
    return fact;
}

int main () {
    int num;
    cout << "Enter the number: ";
    cin >> num;

    cout <<" Is number prime (0→False & 1→True): " << prime (num) <<endl;
    cout << "Factorial: " << Factorial (num) << endl;
    return 0;
}
```

Q) Reverse a number n using function, Constraints

$-5000 \le n \le 5000$

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```
#include <iostream>
using namespace std;

int reverse (int num) {
        int reverse = 0 , ld;
        while (num != 0) {
                ld = num % 10 ;
                num /= 10 ;
                reverse = reverse * 10 + ld;
        }
        return reverse;
}

int main () {
        int num;
        cout << "Enter the number (-5000 to 5000): ";
        cin >> num;
        if (num > 5000 || num < -5000) {
                cout << "Input not in the given range." << endl;
                return 0;
        }
        cout << "Reversed Value: " << reverse(num) << endl;
        return 0;
}
```

Q) Input three numbers a, b, and c. Put the value of a into b, put value of b into c and put value of c into a. Do it using function.

Pass by Reference using &

```
#include <iostream>
using namespace std;
void swap (int &a , int &b , int &c) {
        int temp = a;
        a = c ;
        c = b;
        b = temp;
}
```

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```
int main (){
    int a, b, c;
    cout << " Enter the three numbers: ";
    cin >> a >> b >> c;
    cout << "Before Swap:\n a:" << a <<"\nb:" << b <<"\nC:" << c << endl;
    swap (a,b,c);
    cout << "After Swap:\n a:" << a <<"\nb:" << b<< "\nc:" << c << endl;
    return 0;
}
```

Q) Swap two numbers a, b without using extra variable. Range of
-10 000 ≤ a, b ≤ 100,000

```
#include <iostream>
using namespace std;
void swap (int &a , int &b) {
        a= a+b;
        b = a-b;
        a =a-b;
    }
int main () {
    int a, b;
    cout << " Enter the two numbers: ";
    cin >> a >> b;
    if (a > 100000 || b > 100000 || a <-10000 || b < -10000) {
            cout << " Overflow!" << endl;
            return 0;
        }
    swap (a,b);
    cout << "A: " << a << "B: " << b << endl;
    return 0;
}
```

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